

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

First Steps to Understand Functional Languages



Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations. Talk about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Can Do Statements:

- I can talk about functional languages.
- I can explain what functional languages are.
- I can describe briefly some functional languages.

Ask your partner:

- What does a computer language consist of?
- What types of programming languages do you know?





Let's Explore!

Juan is learning about functional languages with his tutor.



Instructions: Read the text below.

Juan: Good morning, I am looking for one of the tutors, her name is Juliana.

Juliana: That's me. I'm helping you with the tutoring session for today, which is about functional languages, right?

Juan: Yes, it's about that topic. I don't think I'm ready for the exam yet.

Juliana: That's ok, let's start. A functional language is an artificial language that a computer or any other programmable machine executes. In other words, a functional language is a computer language.

Juan: OK, it seems you are using familiar words to explain the topic. The teacher always uses complex vocabulary.

Juliana: That's why tutoring helps students most of the times. We are implementing one rule: the simpler, the better. Let's go back to the topic. In today's world, programmers always use functional languages, which are programming languages based on mathematical functions in all their structures. These mathematical functions possess a mathematical schema derived from lambda calculus and combinatory logic.

Juan: Why are they always using mathematical functions?

Juliana: Well, by using mathematical functions, functional languages become real. In other words, they are readable. This has to do with first class functions; it means that, as you are passing functions as paradigms, you receive them as values. That's why they are real.

Juan: I understand. Please, give me an example of a functional language.

Juliana: Well, one of the most common functional languages that exist is the one programmers are using in applications and program development. Basically, it refers to programs you are working with on a regular basis. For instance, the internet browser you always use. These browsers have a functional language or computer languages such as C, C++, D, Java, and Visual Basic. Another example is the internet and web page development; programmers usually use the same functional languages such as HDML, HTML, Java, JavaScript and Python.



Let's Practice!

Things to Know About Functional Languages

Instructions: Write the words to correctly complete the sentences. Use words from the box below.

real / difficult / mathematical

1. In functional languages, problems can appear in a _____ way.
2. A functional language always becomes _____ by using mathematical functions.
3. Some people think that functional languages are _____ to use.



Let's Practice!

Why are functional languages useful?

Instructions: Select the correct answer.

1. Functional languages are _____ mathematical, while imperative languages are grammatical.
Hint: 100% of the times.
 - a. never
 - b. always
 - c. rarely
 - d. seldom
2. People _____ use LISP to arrange functions.
Hint: 90% of the times.
 - a. never
 - b. almost never
 - c. usually
 - d. rarely
3. Programmers _____ use the most common functional languages for applications and programs development.
Hint: 50% of the times.
 - a. sometimes
 - b. never
 - c. always
 - d. almost never
4. Functional languages _____ use grammar functions.
Hint: 0% of the times.
 - a. sometimes
 - b. never
 - c. always
 - d. almost never
5. Nowadays, programmers _____ use functional languages.
Hint: 100% of the times.
 - a. sometimes
 - b. never
 - c. always
 - d. almost never





Extension Activity

Review what you've learned!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about functional languages
- Choose a functional language and describe it on an email using the present continuous.
- Describe it briefly to any of your partners.

Self-Assessment

Yes

No

- I talk about functional languages.
- I write a short email describing a functional language.
- I speak briefly about a functional language.

☐	☐
☐	☐
☐	☐



Target Words

- **functional language** (noun): artificial language for machinery
e.g. In other words, a functional language is a computer language.
- **schema** (noun): synonym of framework
e.g. These mathematical functions possess a mathematical schema derived from Lambda calculus and combinatory logic.
- **readable** (adjective): easy to be read
e.g. Well, by using mathematical functions, functional languages become real. In other words, they are readable.
- **internet browser** (noun): application that allows you to surf on the internet
e.g. For instance the internet browser you always use.
- **programmable** (adjective): editable in its functions
e.g. A functional language is an artificial language that a computer or any other programmable machine executes.
- **machine** (noun): device used for a specific purposes
e.g. A functional language is an artificial language that a computer or any other programmable machine executes.



Learning Points

Adjectives to Talk About Functional Languages

There are some **specific adjectives** to describe functional languages.

Adjectives always describe people, places, and things. There are some **specific adjectives** that are used to describe functional languages. Basically, they describe some of the characteristics these functional languages have. As a reference, you can use: **free**, **difficult**, **mathematical**, **functional**, **readable**, and **real**.

Adjective	Meaning	Example
Free	Something that doesn't cost anything.	<i>This is a free software.</i>
Difficult	Something hard to achieve or understand.	<i>Functional languages aren'tdifficult.</i>
Mathematical	Related to math.	<i>This app includes mathematical procedures.</i>

Example

- *Well, by using mathematical functions, functional languages become real. In other words, they are readable.*

Present Continuous to Talk About Current Actions

You use the **present continuous** to talk about an action that is happening in a specific time in the present.

To create a statement using the present continuous, you need to use the verb **to be** (am-is-are) and a verb in its **-ing** form to show that the action is happening at a specific moment in the present.

Grammatical Structure:

Pronoun + verb **to be** + verb (**ing**) + complement

Example:

*We **are using** functional languages to use different paradigms as well.*

*Sheila **is using** a new software on her computer.*





Learning Points

Examples

- *I'm helping you with the tutoring session for today, which is about functional languages, right?*
- *Ok, it seems you are using familiar words to explain the topic.*
- *This has to do with first class functions; it means that, as you are passing functions as paradigms, you receive them as values.*

What Is a Language Programming?

There is a specific language programmers use to develop software, scripts, and instructions for computers to execute just by clicking ENTER.

The use of these different languages have helped the computer world to evolve. There are many different fields where you may find language programming.

In the following chart, you will find some examples of language programming.

Application and Program development	Java
Artificial Intelligence development	AIML
Database development	DBASE – FoxPro
Game development	DarkBASIC – Java
Computer drivers or other hardware interface development	Assembly – C
Internet and web page development	HDML- HTML – JavaScript
Script development	awk- Python